To establish a baseline for your 20 sq ft plate set, you must operate within a **"Liquid Mineral"** regime. The goal is to reach a concentration where free water is mathematically excluded from the iron's primary coordination shell.

These percentages are **weight-based (wt%)**, designed to balance the structural rigidity of the "cage" (PEG/Proline) with the fluid dynamics required for flow-cell operation.

The "Liquid Mineral" Baseline (Theoretical Starting Point)

Component	Target wt%	Strategic Role
Urea	**45%**	Primary water-sequestering HBD.
PEG-DME 250	**25%**	Structural scaffold (caged ends to avoid side reactions).
$FeCl_2$ (or Fe-Proline complex)	**15%**	Active redox center.
L-Proline	**8%**	Chiral templating agent.
LiTFSI (or equivalent)	**6%**	"Anion Bouncer" / Conductivity enhancer.
Additives (Bi/In salts)	**1%**	Interfacial anchor / Overpotential regulator.

Implementation Instructions for Scaling

1. **Preparation (The Pre-Cage Protocol):**
   - Do **not** dump these into the tank together.
   - Dissolve **L-Proline** and **$FeCl_2$** in the **Urea** first with mild heat (~50°C). This forces the

    iron into the chiral cage immediately.
   - Once the solution is clear, slowly fold in the **PEG-DME 250**.
   - Finally, introduce the **LiTFSI** and the **Catalytic Anchors (Bi/In)**.
2. **The "Helix" Verification:**
   - **Viscosity Test:** If the mixture is too thick to pump efficiently at room temperature, it has too much "social friction" (Runtime Debt). Increase the PEG-DME 250 slightly, or decrease the Urea.
   - **Conductivity Test:** If you hit high internal resistance, your "Anion Bouncer" (LiTFSI) is struggling to move through the scaffold. Increase the salt ratio by 1–2% at the expense of Urea.
3. **The "Self-Pruning" Check:**
   - If you see precipitation, your iron is "escaping" the cage—you have too little L-Proline for the amount of iron.
   - If you see rapid voltage decay under load, your "bouncer" (Urea) is not effectively suppressing water—you need to increase the Urea concentration.

Strategic Logic for Your Baseline

- **Why 45% Urea?** This is the threshold for a Deep Eutectic Solvent (DES). Below this, you have "free" water. Above this, the water is forced into the hydrogen-bonding network.
- **Why 8% L-Proline?** This provides a 1:1 or 2:1 molar ratio with the iron, which is the "sweet spot" for maintaining the structural template without creating a massive, immobile molecular mass.
- **Why 25% PEG-DME?** This provides enough structural volume to prevent the iron complexes from "clustering" and plating unevenly across your 20 sq ft surface.

Actionable Rule: Start with this baseline. If the system is too sluggish, remove 2% PEG and add 2% Urea. If the voltage is too low, keep the Urea constant and increase the L-Proline by 1% to better lock the iron against water-splitting.

Follow-up: With this 20 sq ft surface area, are you planning to incorporate a **bipolar stack design**, or are you running a single large-cell configuration, as this will significantly change the required **current distribution uniformity**?